

Autonomous Driving Capstone — STAR Write-up

Project: Course 7 Capstone Project 1 — Autonomous Driving

Client framing: AV / ITS vehicle perception + Tesla Autopilot safety analytics

Stack: Python, TensorFlow/Keras (MobileNetV2), OpenCV, pandas, scikit-learn, seaborn

Data: Road-scene images with bounding boxes · Tesla Deaths public CSV

Situation

Autonomous vehicles and intelligent transport systems need reliable **on-road object understanding** — not just “is something there,” but **what** it is and **where** it sits in the frame. Separately, Tesla’s Full Self-Driving / Autopilot narrative makes **fatal-incident patterns** a high-visibility public-safety and product-risk topic.

This capstone combined both:

1. **Part 1** — deep-learning vehicle-type recognition and localization visuals on annotated driving frames
2. **Part 2** — structured EDA of Tesla fatality events (calendar, geography, victims, models, verified Autopilot deaths)

Task

Part 1 — Object detection / localization

1. Create parent/child folders for custom model training
2. Prepare the dataset (filter labels to images that exist on disk)
3. Build a CNN of choice for detection / classification + localization presentation
4. Evaluate with clear metrics
5. Run inferences on sample images with annotated boxes

Part 2 — Tesla Deaths

1. Preliminary inspection (dtypes, missing, duplicates) and drop irrelevant columns
2. EDA covering events by date/year/day/state/country; death aspects; model mix; verified Autopilot deaths

Action

Part 1

- Documented training layout under `data/part1` (`Images/`, `labels.csv`, `crops/`) plus `models/` and `outputs/`.

- Loaded headerless `labels.csv` (`img, cls, xmin, ymin, xmax, ymax`), padded IDs to 8 digits, and **filtered to 5,626 on-disk images (~17.9k boxes)**.
- Dropped tiny boxes (<24px) and ambiguous catch-all labels (`motorized\_vehicle`, `non-motorized\_vehicle`) that collide with car/pickup, then built a **stratified ~3.6k-crop subset across 9 classes**.
- Trained a **MobileNetV2 (ImageNet) transfer-learning crop classifier** (frozen head → light fine-tune) with mild class weights and augmentation.
- Added a **multi-output MobileNet** (class + normalized bbox) on the largest object per frame as a localization demo (~71% val class accuracy).
- Evaluated with accuracy, classification report, and confusion matrix; drew predicted labels on GT boxes and saved annotated inferences.

Part 2

- Cleaned messy spaced column names; treated `` as missing; coerced numerics.
- Dropped URL/Unnamed columns, free-text `Description`, `Source`/`Note`, deceased-name PII, and **footer aggregate rows**.
- Produced charts for calendar/geography, death aspects, model distribution, and verified Autopilot deaths.

Result

Part 1 — perception model

Metric	Value
Crop classifier val accuracy (9-class)	**68.6%**
Coarse-group accuracy (passenger / large / VRU)	**89.9%**
Multi-output val class accuracy	~71.2%
Train / val crops	2,900 / 725

Ambiguous catch-all labels were the main noise source on the full 11-class taxonomy; after dropping them, fine-grain accuracy approached the 70% target and coarse grouping exceeded it. Annotated sample frames live in `outputs/inference\_\*.jpg`. Saved Keras models: `models/crop\_classifier\_final.keras`, `models/class\_bbox\_multioutput.keras`.

Part 2 — Tesla safety EDA (highlights)

- **293** cleaned fatal events · **352** total deaths · mean **~1.20 deaths/event**
- Tesla **driver** died in **~40%** of events; driver *or* occupant death in **~47%**
- **Cyclist/pedestrian** involvement: **~15%** of events
- Collision with **other vehicle(s)**: **~38%**
- Models mostly **unspecified**, then **S / 3 / X / Y**
- **Verified Autopilot death events**: 16 (19 verified deaths total) vs **34** Autopilot-*claimed* events — claims exceed verified counts

Artifacts

- Executed notebook: `autonomous\_driving.ipynb`

- Charts & inferences: `outputs/`
- Models: `models/`
- LMS package: `lms\_upload/` (write-up PDF, screenshots zip, notebook)

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Quantified results (from executed run)

- Crop classifier val accuracy: **68.55%**
- Coarse-group val accuracy: **89.93%**
- Train/val crops: 2900/725 · classes: 9
- Tesla events analyzed: 293
- Tesla driver death events: 117 (39.9%)
- Other-vehicle collisions: 110 (37.5%)
- Verified Autopilot death events: 16